

3.5 SUMMARY

In this chapter we discussed least squares estimation in a couple of different contexts. First we derived a method for estimating a constant vector on the basis of several noisy measurements of that vector. In fact, the measurements do not have to be direct measurements of the constant vector, but they can be measurements of some linear combination of the elements of the constant vector. In addition, the noise associated with each measurement does not have to be the same. The least squares estimation technique that we derived assumed that the measurement noise is zero-mean and white (uncorrelated with itself from one time step to the next), and that we know the variance of the measurement noise. We then extended our least squares estimator to a recursive formulation, wherein the computational effort remains the same at each time step regardless of the total number of measurements that we have processed. Least squares estimation of a constant vector forms a large part of the foundation for the Kalman filter, which we will derive later in this book.

In Section 3.4, we took a brief segue into Wiener filtering, which is a method of estimating a time-varying signal that is corrupted by noise. The Wiener filter is based on frequency domain analyses, whereas the Kalman filter that we derive later is based on time domain analyses. Nevertheless, both filters are optimal under their own assumptions. Some problems are solvable by both the Wiener and Kalman filter methods, in which case both methods give the same result.

PROBLEMS

Written exercises

3.1 In Equation (3.6) we computed the partial derivative of our cost function with respect to our estimate and set the result equal to 0 to solve for the optimal estimate. However, the solution minimizes the cost function only if the second derivative of the cost function with respect to the estimate is positive semidefinite. Find the second derivative of the cost function and show that it is positive semidefinite.

3.2 Prove that the matrix P_k that is computed from Equation (3.25) will always be positive definite if P_{k-1} and R_k are positive definite.

3.3 Consider the recursive least squares estimator of Equations (3.28)-(3.30). If zero information about the initial state is available, then $P_0 = \infty I$. Suppose that you have a system like this with $H_k = 1$. What will be the values of K_1 and P_1 ?

3.4 Consider a battery with a completely unknown voltage ($P_0 = \infty$). Two independent measurements of the voltage are taken to estimate the voltage, the first with a variance of 1, and the second with a variance of 4.

- a) Write the weighted least squares voltage estimate in terms of the two measurements y_1 and y_2 .
- b) If weighted least squares is used to estimate the voltage, what is the variance of voltage estimate after the first measurement? What is the variance of the voltage estimate after the second measurement?

- c) If the voltage is estimated as $(y_1 + y_2)/2$, an unweighted average of the measurements, what is the variance of the voltage estimate?

3.5 Consider a battery whose voltage is a random variable with a variance of 1. Two independent measurements of the voltage are taken to estimate the voltage, the first with a variance of 1, and the second with a variance of 4.

- Write the weighted least squares voltage estimate in terms of the initial estimate $\hat{x}_0$ and the two measurements y_1 and y_2 .
- If weighted least squares is used to estimate the voltage, what is the variance of voltage estimate after the first measurement? What is the variance of the voltage estimate after the second measurement?

3.6 Suppose that $\{x_1, x_2, \dots, x_n\}$ is a set of random variables, each with mean $\bar{x}$ and variance σ^2 . Further suppose that $E[(x_i - \bar{x})(x_j - \bar{x})] = 0$ for $i \neq j$. We estimate $\bar{x}$ and σ^2 as follows.

$$\hat{\bar{x}} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\bar{x}})^2$$

- Is $\hat{\bar{x}}$ an unbiased estimate of $\bar{x}$? That is, is $E(\hat{\bar{x}}) = \bar{x}$?
- Find $E(x_i x_j)$ in terms of $\bar{x}$ and σ^2 for both $i = j$ and $i \neq j$.
- Is $\hat{\sigma}^2$ an unbiased estimate of σ^2 ? That is, is $E(\hat{\sigma}^2) = \sigma^2$? If not, how should we change $\hat{\sigma}^2$ to make it an unbiased estimate of σ^2 ?

3.7 Suppose a scalar signal has the values 1, 2, and 3. Consider three different estimates of this time-varying signal. The first estimate is 3, 4, 1. The second estimate is 1, 2, 6. The third estimate is 5, 6, 7. Create a table showing the RMS value, average absolute error, and standard deviation of the error of each estimate. Which estimate results in the error with the smallest RMS value? Which estimate results in the error with the smallest infinity-norm? Which estimate gives the error with the smallest standard deviation? Which estimate do you think is best from an intuitive point of view? Which estimate do you think is worst from an intuitive point of view?

3.8 Suppose a random variable x has the pdf $f(x)$ given in Figure 3.3.

- a) x can be estimated by taking the median of its pdf. That is, $\hat{x}$ is the solution to the equation

$$\int_{-\infty}^{\hat{x}} f(x) dx = \int_{\hat{x}}^{\infty} f(x) dx$$

Find the median estimate of x .

- b) x can be estimated by taking the mode of its pdf. That is,

$$\hat{x} = \arg \max f(x)$$

Find the mode estimate of x .

- c) x can be estimated by computing its mean. That is,

$$\hat{x} = \int_{-\infty}^{\infty} x f(x) dx$$

Find the mean of x .

- d) x can be estimated by computing the minimax value. That is,

$$\hat{x} = \min_x \max |x - \hat{x}|$$

Find the minimax estimate of x .

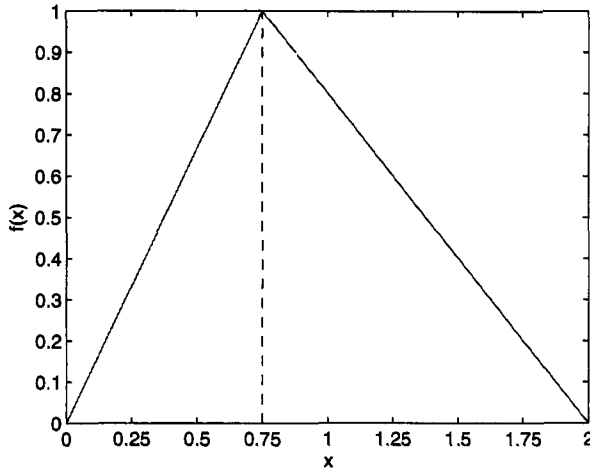


Figure 3.3 pdf for Problem 3.8.

3.9 Suppose you are responsible for increasing the tracking accuracy of a radar system. You presently have a radar that has a measurement variance of 10. For equal cost you could either: (a) optimally combine the present radar system with a new radar system that has a measurement variance of 6; or, (b) optimally combine the present radar system with two new radar systems that both have the same performance as the original system [May79]. Which would you propose to do? Why?

3.10 Consider the differential equation

$$\dot{x} + 3x = u$$

If the input $u(t)$ is an impulse, there are two solutions $x(t)$ that satisfy the differential equation. One solution is causal and stable, the other solution is anticausal and unstable. Find the two solutions.

3.11 Suppose a signal $x(t)$ with power spectral density

$$S_x(s) = \frac{1 - s^2}{s^4 - 5s^2 + 4}$$

is corrupted with additive white noise $v(t)$ with a power spectral density $S_v(s) = 1$.

- Find the optimal noncausal Wiener filter to extract the signal from the noise corrupted signal.
- Find the optimal causal Wiener filter to extract the signal from the noise corrupted signal.

3.12 A system has the transfer function

$$G(s) = \frac{1}{s-3}$$

If the input is an impulse, there are two solutions for the output $x(t)$ that satisfy the transfer function. One solution is causal and unstable, the other solution is anticausal and stable. Find the two solutions.

Computer exercises

3.13 The production of steel in the United States between 1946 and 1956 was 66.6, 84.9, 88.6, 78.0, 96.8, 105.2, 93.2, 111.6, 88.3, 117.0, and 115.2 million tons [Sor80]. Find the least squares fit to these data using (a) linear curve fit; (b) quadratic curve fit; (c) cubic curve fit; (d) quartic curve fit. For each case give the following: (1) a plot of the original data along with the least squares curve; (2) the RMS error of the least squares curve; (3) the prediction of steel production in 1957.

3.14 Implement the Wiener filters for the three examples given in Section 3.4 and verify the results shown in Section 3.4.5. Hint: Example 8.6 shows that if $\dot{x} = -x + w$ where $w(t)$ is white noise with a variance of $Q_c = 2$, then

$$S_x(\omega) = \frac{2}{\omega^2 + 1}$$

From Sections 1.4 and 8.1 we see that this system can be simulated as

$$\begin{aligned} x(t + \Delta t) &= e^{-\Delta t} x(t) + w(t) \sqrt{Q_c \Delta t} \\ y(t) &= x(t) + v(t) \sqrt{R_c / \Delta t} \end{aligned}$$

where $w(t)$ and $v(t)$ are independent zero-mean, unity variance random variables.

